Cpts 317. 10/17.

Prob 3. Show $L = \{0^n 1^m : n \le m\}$ is not regular.

Proof. use pumping lemma by taking $z = 0^n 1^n$

where n is the ~~lem~~ number in the lemma.

(when you pump in the last step, take $i=0$).

Proof (Prob 3). Assume it's regular. Then, according to PL., Let n be the number in the lemma.

Pick $z = 0^n 1^n \in L$. Assume $z = uvw$ s.t. $|uv| \leq n$, $|v| \geq 1$.

$$z = \overbrace{0 \cdots 0}^{n} \overbrace{1 \cdots 1}^{n}$$

$$z = u \quad v \quad \leftarrow \quad w \quad \rightarrow$$

Clearly, u and v contain 0's only. Take $i = m = n$, from the lemma, $uv^{m=n}w \in L$. However this is a contradiction since in $uv^{m=n}w = \underbrace{uv \cdots v}_{m \text{ times}} w$, there are more 0's than 1's.

Clearly, u and v contain 0's only. Take $i = n$. From the lemma, $uv^n w \in L$. However, this is a contradiction since in $uv^n w = u\underbrace{v \cdots v}_{n} w$ there are more 0's than 1's.

$\blacksquare$

" you can also take $i = 2$. (repeat v for
" twice), it's also okay.

#4. Show that $L = \{ x x^R x : x \in (a+b)^* \}$ is not regular.

Proof. (PL). Take $z = a^n b^n b^n a^n a^n b^n$, ...

take $i=0$ ($i=2$ is okay), ...

#5.

(1). not reg.

(2). reg. $(0000(00)^*)$.

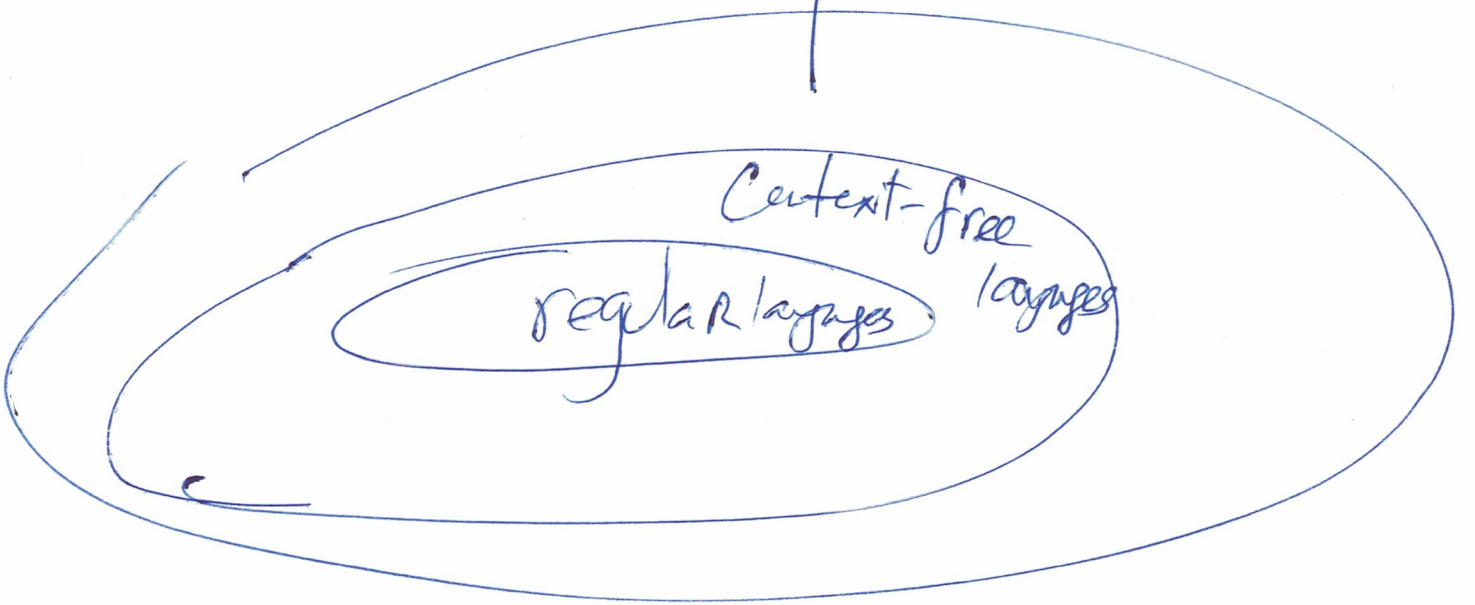
(3). reg. $(0+1)^*$.

(4). not reg. $(i=0)$.

Today: Context-free grammars.

Chomsky.

language hierachy



$L = \{0^n 1^n : n \geq 1\}$ is not reg.

$$\underbrace{00\cdots0}_{n}\ \underbrace{11\cdots1}_{n}$$

$$S \longrightarrow 0S1 \mid 01$$ = grammar.

$$S \Rightarrow 0S1$$
$$\Rightarrow 00S11$$
$$\vdots$$
$$\Rightarrow 0\cdots0S1\cdots1$$
$$\Rightarrow \underbrace{0\cdots00}_{n}\ \underbrace{1\cdots11}_{n}$$

Exaple. $L = \{(101)^n(110)^n : n \geq 0\}$

c.f. grammar:

$$S \longrightarrow 101S110 \mid \Lambda$$

Try: $101101110110 \in L$.

$$S \Rightarrow 101S110$$
$$\Rightarrow 101\,101S110\,110$$
$$\Rightarrow 101\,101\,110\,110 \in L.$$

Exaple. $L = \{ 1^n 0^n 1^m 0^m : n \geq 0, m \geq 0 \}$.

Grammar:

$$S \longrightarrow AB$$

$$A \longrightarrow 1A0 \mid \Lambda$$

$$B \longrightarrow 1B0 \mid \Lambda.$$

Check: $110010 \in L$.

$$\begin{aligned} S &\Longrightarrow AB \\ &\Longrightarrow 1A0B \\ &\Longrightarrow 11A00B \\ &\Longrightarrow 1100B \\ &\Longrightarrow 11001B0 \Longrightarrow 110010 \in L. \end{aligned}$$

"Context-free": replacement is independent of the symbols around the replaced symbol.

~~Today's Chandge:~~

① English is c.f. or not?

② C-program has a gramar. Is it c.f. or not?

$\langle Expr \rangle ::= \langle Expr \rangle + \langle Expr \rangle \mid \langle Expr \rangle - \langle Expr \rangle \mid \ldots$

Backus-Noar form.

We do have context-sensitive grammars (which are more powerful than c.f. grammars). Example:

$$AB \longrightarrow AAB.$$